

'atop': 'atop' provides a more detailed view of the processes running on your system compared to the 'top' command. It allows you to view historical data about the processes running on your system, as well as real-time data that can help in identifying trends and patterns in the behaviour of your system.

'glances': This is a multi-platform, user-friendly tool that provides a comprehensive view of system performance, including CPU usage, memory usage, disk usage, and network activity.

Each of these tools has its own strengths and weaknesses. 'top' is a good choice for a simple and lightweight tool that provides real-time information about the processes running on your system and is included in most Linux distributions. 'htop' is a more advanced tool with a more user-friendly interface. Its additional features allow you to sort processes by various criteria, search for specific processes, and view detailed information about each process.

If you need a tool that provides historical data about the processes running on your system, as well as real-time

data, 'atop' is more suitable. If you need a multi-platform, user-friendly tool that provides a comprehensive view of the performance of your system, 'glances' may be better as it gives all the information in one place. The best tool for monitoring the processes running on your Linux system depends on your specific needs and preferences.

There are a few other commands that may be helpful in certain situations.

'pstree': This command displays a tree-like representation of all the processes currently running on the system, and is useful for visualising the relationships between parent and child processes.

'pidof': This command helps you find the process ID (PID) of a running program by name. For example, if you want to find the PID of the chrome process, you can run the following command:

```
$pidof chrome
```

'pgrep': This command enables you to search for processes based on their attributes, such as name or owner. For example, if you want to find all processes owned by the user ravi, you can run the following command:

```
$pgrep -u ravi
```

This can help automate the task of finding and manipulating a process or a group of processes.

These commands may be helpful in certain situations and learning them can make you more effective at managing processes in Linux.

In this tutorial, we have explored various commands that can monitor the processes running on a Linux system. We have discussed the 'ps', 'kill', 'nice', 'renice', 'bg', 'fg', and 'top' commands and their most frequently used options, the commands 'nice' and 'renice' for adjusting the priorities of processes. We have introduced other commands like 'htop', 'atop', and 'glances' that can achieve similar results. We have briefly mentioned a few less frequently used commands that are useful.

By mastering these commands and their options, you can gain valuable insights into the processes running on your Linux system and manage system resources more effectively. **END** 

Table 2: Process management commands and their functions

Command	Description
<i>kill</i>	Terminates a process
<i>kill -l</i>	Prints a list of signal names that can be used with the 'kill' command
<i>kill -9</i>	Forcibly ends a process
<i>killall</i>	Kills processes by name
<i>killall -u ravi</i>	Kills all processes owned by a specific user 'ravi'
<i>bg</i>	Resumes a suspended process in the background
<i>fg</i>	Brings a job running in the background to the foreground
<i>jobs</i>	Displays the status of jobs currently running or suspended in the background
<i>nice</i>	Runs a program with changed scheduling priority
<i>renice</i>	Alters priority of running processes
<i>top</i>	Displays currently executing Linux processes interactively in real-time
<i>htop</i>	An interactive and user-friendly process monitoring tool for Linux
<i>atop</i>	A real-time system monitor that displays an anatomised view of the system performance
<i>glances</i>	A cross-platform tool that gives a comprehensive view of the performance of your system, in a user-friendly way
<i>pstree</i>	Displays a tree of processes
<i>pidof</i>	Finds the process ID of a running program
<i>pgrep</i>	Looks up or signals processes based on name and other attributes

Reference

- Linux man pages

By: Sathyanarayanan S.

The author is working with Sri Sathya Sai University for Human Excellence, Gulbarga. He has more than 30 years of experience in systems management and teaching IT courses. He is an enthusiastic promoter of FOSS.